Humid adiabatic atmosphere --- Solution

Y22 (2022) — English translation

A1 **Find the air temperature $T_{\text{в}}$ at a height h above the Earth's surface. Express your answer in terms of T_0, g, $\mu_{\text{в}}$, R and the adiabatic exponent $\gamma_{\text{в}}$.** **1.00**

Let's consider moving a portion of air to a height h:

$$P_1 dV_1 - P_2 dV_2 = \frac{C_V dm}{\mu_{\text{в}}}(T - T_0) + dmgh$$

from where:

Answer:

$$T = T_0 - \frac{\mu_{\text{в}} gh}{C_P} = T_0 - \frac{(\gamma_{\text{в}} - 1)\mu_{\text{в}} gh}{\gamma_{\text{в}} R}$$

A2 **Find the dependence of water vapor pressure on temperature $p_{\text{п}}(T)$. Express your answer in terms of φ_0, $p_{\text{п}0}$, T_0, $\mu_{\text{п}}$, $\mu_{\text{в}}$, $\gamma_{\text{в}}$ and T.** **1.00**

From the condition of steam equilibrium we obtain:

$$dP_{\text{п}} = -\rho_{\text{п}} g dh$$

откуда:

$$\frac{dP_{\text{п}}}{P_{\text{п}}} = \frac{\mu_{\text{п}}\gamma_{\text{в}}}{\mu_{\text{в}}(\gamma_{\text{в}} - 1)} \frac{dT}{T}$$

Integrating, we find:

Answer:

$$P_{\text{п}} = \varphi_0 P_{\text{п}0} \left(\frac{T}{T_0}\right)^{\frac{\mu_{\text{п}}\gamma_{\text{в}}}{\mu_{\text{в}}(\gamma_{\text{в}} - 1)}}$$

A3 **Using the graph of $p_{\text{н.п}}(T)$, find for $\varphi_0 = 65\%$ and $T_0 = 287$ К the height h above the Earth's surface at which the vapor becomes saturated.** **0.50**

Solving the equations graphically, we find:

$$T_{\text{нас}} \approx 6{,}6^{\circ}C$$

Then:

Answer:

$$h = \frac{\gamma_в R(T_0 - T_{нас})}{(\gamma_в - 1)\mu_в g} \approx 757 \text{ м}$$

B1 **The specific heat of vaporization depends on temperature T as follows:** **1.00**

$$\lambda(T) = \lambda_n + k(T - T_n)$$

Найдите k. Ответ выразите через c, R, $\mu_п$ и показатель адиабаты $\gamma_п$. $Note:$ если вы не можете решить этот пункт, то в дальнейшем считайте, что $\lambda = \lambda_n = const$. This does not lead to loss of points.

The amount of heat required to evaporate mass m is equal to:

$$Q = U_п - U_ж + A$$

Поскольку плотностью воды можно пренебречь:

$$A = pV = \frac{mRT}{\mu}$$

Поскольку жидкость слабо сжимаема, количество теплоты, необходимое для её нагревания в любом процессе, с хорошей точностью пропорционально изменению её внутренней энергии:

$$U_ж(T_2) - U_ж(T_1) = cm(T_2 - T_1)$$

тогда

$$U_ж(T) = cmT + mC$$

где C - некоторая постоянная величина Тогда:

$$Q = mC + m\left(\frac{\gamma_п R}{\mu_п(\gamma_п - 1)} - c\right)T = \lambda(T)m$$

Определяя C из начальных условий, находим:

$$\lambda(T) = \lambda_n + \left(c - \frac{\gamma_п R}{(\gamma_п - 1)\mu_п}\right)(T_n - T)$$

We get the answer:

Answer:

$$k = \frac{\gamma_п R}{(\gamma_п - 1)\mu_п} - c$$

B2 **Show that when the released air mass moves, the amount of heat received by the system enclosed in its volume can be represented as follows:** **0.50**

$$\delta Q = m_в \left(\frac{\gamma_в R}{\mu_в(\gamma_в - 1)} dT + g dz + \lambda d\alpha \right)$$

For the amount of heat supplied to the system we have:

$$\delta Q = \frac{C_V m_в}{\mu_в} + P_в dV_в + \lambda(T) dm_п$$

или же

$$\delta Q = \frac{C_P m_в}{\mu_в} dT - V_в dP_в + \lambda(T) dm_п$$

Из условия равновесия воздуха:

$$dP_в = -\rho_в g dh$$

откуда:

$$\delta Q = m_в \left(\frac{\gamma_в R}{\mu_в(\gamma_в - 1)} dT + g dz + \lambda(T) d\alpha \right)$$

B3 **Find $\frac{d\alpha}{dz}$. Express your answer using $\mu_п$, $\mu_в$, $P_в$, $P_{н.п}$, $\frac{dP_{н.п}}{dT}$, $\frac{dT}{dz}$ and $\frac{dP_в}{dz}$** **0.50**

For α, taking into account the equation of state, we have:

$$\alpha = \frac{m_п}{m_в} = \frac{\mu_п}{\mu_в} \frac{P_п}{P_в}$$

Дифференцируя:

$$\frac{d\alpha}{dz} = \frac{\mu_п}{\mu_в} \left(\frac{1}{P_в} \frac{dP_п}{dz} - \frac{P_п}{P_в^2} \frac{dP_в}{dz} \right)$$

Рассмотрим производную давления пара по высоте как производную сложной функции:

$$\frac{dP_п}{dz} = \frac{dP_п}{dT} \frac{dT}{dz}$$

from where:

Answer:

$$\frac{d\alpha}{dz}=\frac{\mu_п}{\mu_в}\left(\frac{1}{P_в}\frac{dP_п}{dT}\frac{dT}{dz}-\frac{P_п}{P_в^2}\frac{dP_в}{dz}\right)$$

B4 **Find the temperature gradient $\frac{dT}{dz}$. Express your answer in terms of α, g, R, $\gamma_в$, T, $\mu_в$, $\mu_п$ and $\lambda(T)$.** **1.00**

Equating δQ to zero, we find:

$$\frac{dT}{dz}=-\frac{g-\lambda\frac{\mu_п}{\mu_в}\frac{P_п}{P_в^2}\frac{dP_в}{dz}}{\frac{\gamma_в R}{\mu_в(\gamma_в-1)}+\lambda\frac{\mu_п}{\mu_в}\frac{1}{P_в}\frac{dP_п}{dT}}$$

Величину $\frac{dP_п}{dT}$ найдём из уравнения Клапейрона - Клаузиуса:

$$\frac{dP_п}{dT}=\frac{\mu_п\lambda(T)P_п}{RT^2}$$

After substitution we get:

Answer:

$$\frac{dT}{dz}=-\frac{(\gamma_в-1)\mu_в g}{\gamma_в R}\frac{\left(1+\frac{\lambda(T)\mu_в\alpha}{RT}\right)}{\left(1+\frac{\lambda(T)\mu_п\alpha}{RT}\frac{(\gamma_в-1)\lambda(T)\mu_в}{\gamma_в RT}\right)}$$

B5 **Find the temperature T_0 on the Earth's surface.** **1.80**

Let's look at the break point on the graph. At this point $\alpha=\alpha_1$. The reason for the break is the beginning of the condensation process, and therefore for the temperature gradient to the left of the break point the formula from part $A1$ is applicable, and to the right - the formula from part $B4$. Then the value of α_1 can be found from the ratio of the angular coefficients at the break point:

$$k=-\left(\frac{dT}{dz}\right)_{T\to T_1-0}\frac{\gamma_в R}{(\gamma_в-1)\mu_в g}$$

откуда:

$$\alpha_1=\frac{1-k}{\frac{\lambda(T_1)\mu_п}{RT_1}\left(k\frac{(\gamma_в-1)\mu_в g}{\gamma_в R}\frac{\lambda(T_1)}{gT_1}-\frac{\mu_в}{\mu_п}\right)}\approx 1{,}01\cdot 10^{-2}$$

Далее:

$$\alpha_1 = \frac{\mu_{\text{п}} p_{\text{н.п}}(T_1)}{\mu_{\text{в}} p_{\text{в}}}$$

Из уравнения адиабаты для сухого воздуха имеем:

$$p_{\text{в}} = p_0 \left(\frac{T_1}{T_0}\right)^{\frac{\gamma_{\text{в}}}{\gamma_{\text{в}}-1}}$$

Then we get:

Answer:

$$T_0 = T_1 \left(\frac{\alpha_1 \mu_{\text{в}} p_0}{\mu_{\text{п}} p_{\text{н.п}}(T_1)}\right)^{\frac{\gamma_{\text{в}}-1}{\gamma_{\text{в}}}} \approx 309{,}2 \text{ К}$$

B6 **Estimate at what altitude h_2 snow begins to fall.** **0.70**

From the slope of the first section of the graph, we find the height h_1, corresponding to the break in the graph:

$$h_1 = \frac{\gamma_{\text{в}} R(T_0 - T_1)}{(\gamma_{\text{в}} - 1)\mu_{\text{в}} g} \approx 2680 \text{ м}$$

Проведя прямую, соответствующую температуре $T_{\text{pl}} = 273$ K, we find:

Answer:

$$h_2 \approx 4{,}69 \text{ км}$$